

🏠 Employment & Education

Education Level

Bachelor's

Employment Status

Employed

💰 Financial Information

Annual Income (\$)

50000

Monthly Income (\$)

5000

Credit Score

650

Debt to Income Ratio

0.30

🏠 Loan Details

Loan Amount (\$)

10000

Loan Purpose

Business

Grade / Subgrade

A1

Interest Rate (%)

10.00

Loan Term

36 Months

Monthly Installment (\$)

450.00

📄 Credit & Account Info

Open Accounts

5

Total Credit Limit (\$)

40000.00

AI-POWERED RISK ANALYTICS

🏠 AI Loan Intelligence System

Advanced Banking Risk Prediction Engine

Production-grade ML pipeline with multi-model evaluation, real-time risk scoring, and comprehensive financial analytics. Powered by **XGBoost**.

BEST MODEL

XGBoost

Top performer by ROC AUC

BEST ACCURACY

90.28%

Highest across all models

AVG PRECISION

89.42%

Mean of all models

DATASET SIZE

20,000

Training records

PREDICTIONS

0

Session predictions

Predict Loan Status

APPROVAL PROBABILITY

87.32%

RISK LEVEL

● Low

APPLICATION STATUS

APPROVED

✓

CONFIDENCE

High

📊 Risk Meter

Loan Approval Probability

020406080100

87.3%

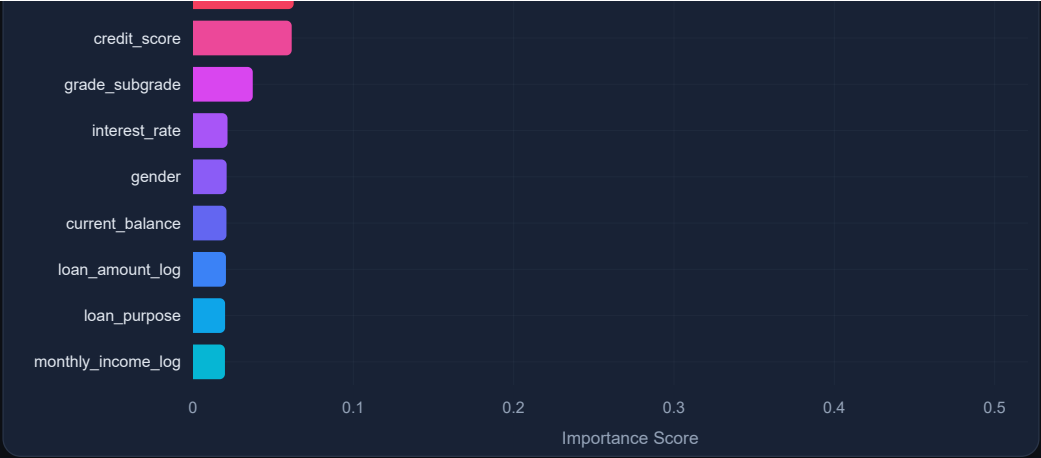
▲37.3

87.32%

📌 AI Feature Importance

employment_status

debt_to_income_ratio



AI Decision Insight

✓ Strong financial profile detected. Low default probability.

AI Loan Assessment Report

Generate & Download AI Report

AI Powered FinTech System | Advanced Machine Learning Project

Machine Learning Model Comparison

DASHBOARD

Performance Metrics Table

TABLE

Model	Accuracy	Precision	Recall	F1 Score	ROC AUC
XGBoost	0.90	0.90	0.99	0.94	0.8
Random Forest	0.90	0.90	0.99	0.94	0.8
Decision Tree	0.90	0.90	0.99	0.94	0.8
Logistic Regression	0.88	0.89	0.97	0.93	0.8

BEST PERFORMING MODEL

PRODUCTION READY

 XGBoost



 Individual Metric Performance

PER-METRIC

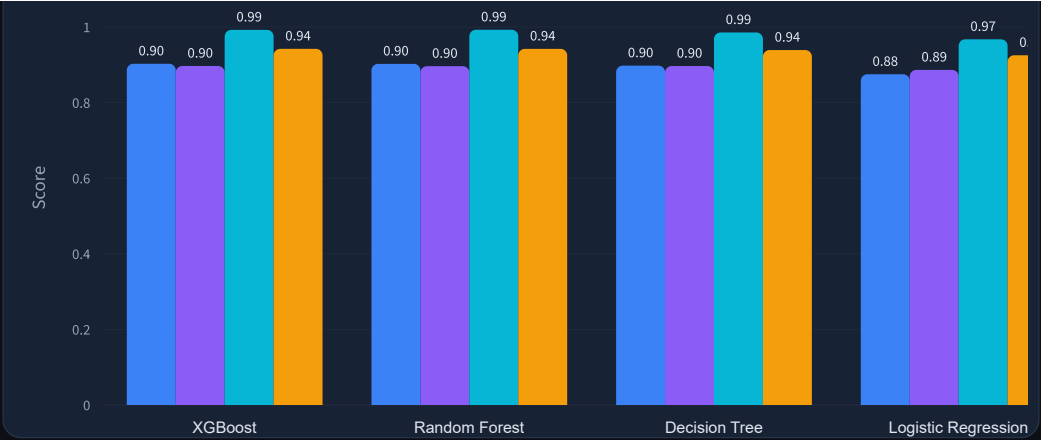


 Model Metrics Comparison

GROUPED

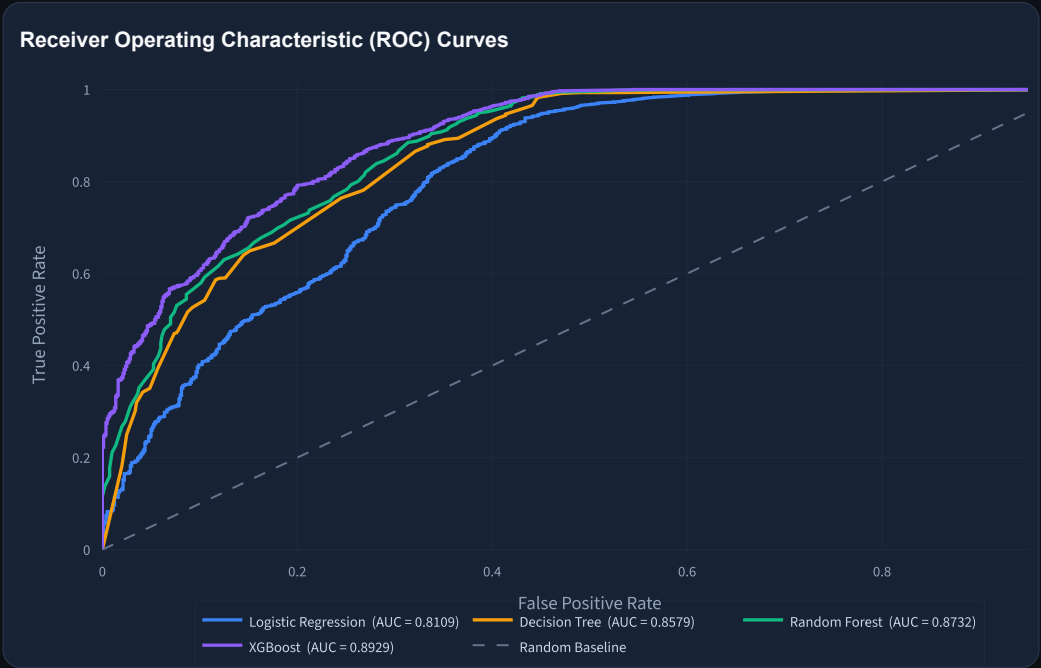
Accuracy / Precision / Recall / F1

 Accuracy  Precision  Recall  F1 Score



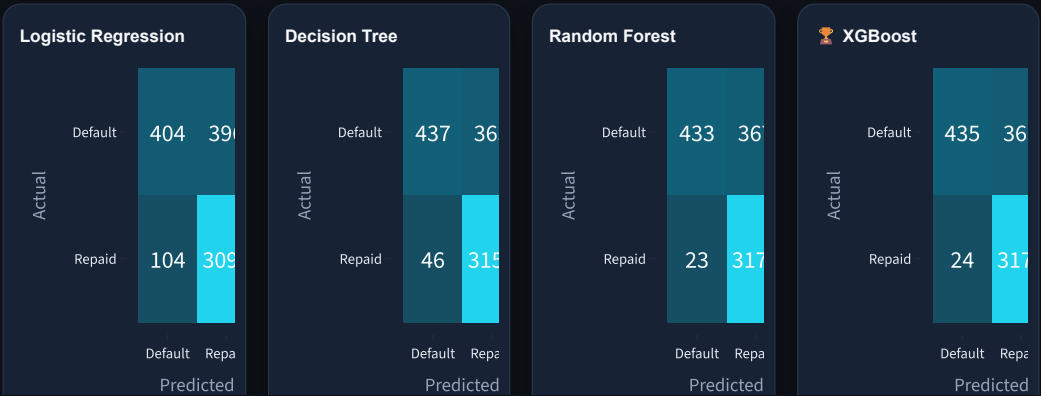
ROC Curve Comparison

AUC



Confusion Matrices

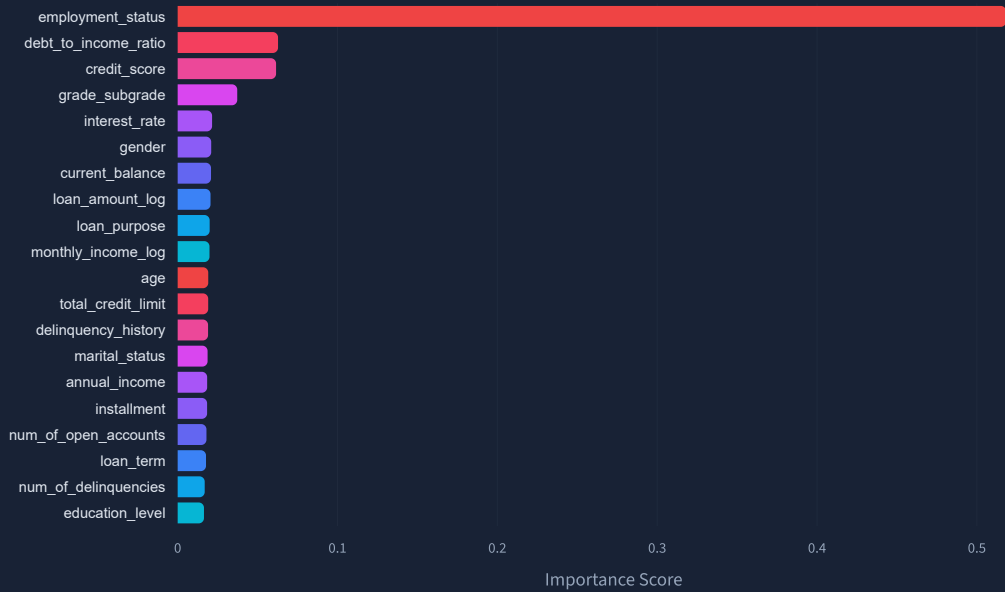
CLASSIFICATION



 Feature Importance — Best Model

EXPLAINABILITY

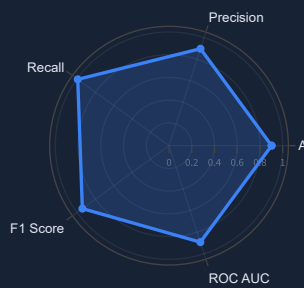
Top 20 Features — XGBoost



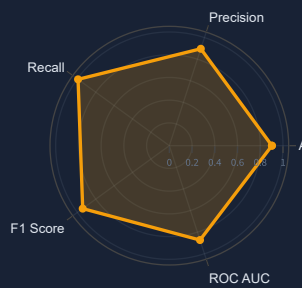
Performance Radar

RADAR

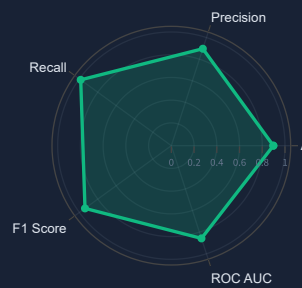
 XGBoost



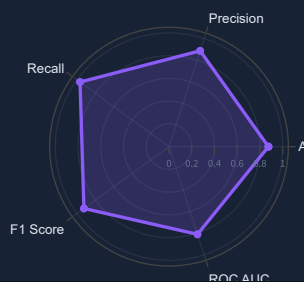
Random Forest



Decision Tree

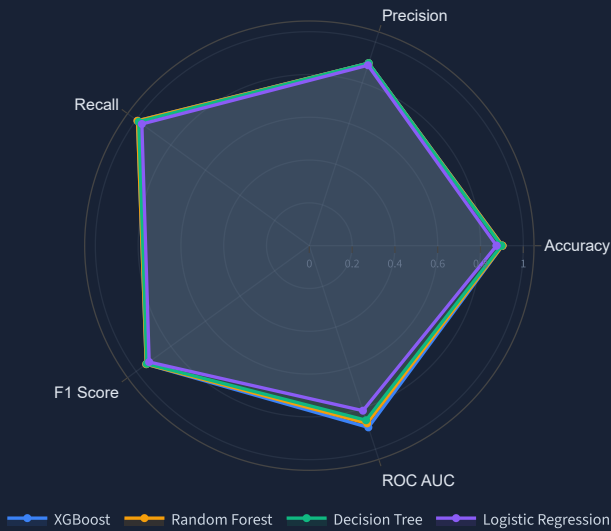


Logistic Regression



Combined Radar Comparison

All Models — Metric Radar



AI Performance Insights

AI ANALYSIS

INSIGHTS FOR ROC AUC

- XGBoost** dominates in **ROC AUC** with a score of **0.8929**.
- Ensemble models average **0.9026** accuracy vs **0.8864** for baseline models — a **1.6pp** improvement.
- Random Forest** leads in **Recall** (0.9928), showing strength in a different dimension.
- XGBoost** is the most stable model (cross-metric std: 0.0423).
- Random Forest** has the highest recall (**0.9928**) — best at catching actual defaults.
- Logistic Regression** underperforms on ROC AUC (0.8109), suggesting lower generalization.

RECOMMENDED PRODUCTION MODEL

XGBoost

- ✓ Highest Accuracy
- ✓ Highest Precision
- ✓ Highest F1 Score
- ✓ Highest ROC AUC
- ✓ Best Generalization

METRIC BREAKDOWN

Accuracy	0.9028 (Rank #1)
Precision	0.8969 (Rank #1)
Recall	0.9925 (Rank #2)
F1 Score	0.9423 (Rank #1)

ROC AUC	0.8929 (Rank #1)
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Export

DOWNLOAD

Download Model Comparison Report (CSV)

> @AI Financial Assistant — Click to open chat